

Answer question1:

Since the healthcare provider needs to store and process amounts of data, which must comply with strict privacy policies and regulations and also desires the flexibility to experiment with new AI-driven diagnostic tools,

so, the most suitable cloud model is : **Hybrid Cloud**

justify the Answer:

This cloud model combines both public and private cloud infrastructure, meeting the organization's requirements as follows:

The private cloud	Since patient data is sensitive and subject to strict regulations, using a private cloud to store and process sensitive data ensures greater security. The private cloud allows the organization to have full control over the data and the security infrastructure
The public cloud	The public cloud can be used to run AI applications and data analytics, while patient data remains protected in the private cloud
the hybrid cloud	Therefore, the hybrid cloud model facilitates seamless integration between private and public clouds, enabling the organization to ensure patient data security while also providing flexibility and access to AI tools.

Answer question2:

A group of developers is working on a new application and wants an environment that allows them to write, test, and deploy code without worrying about managing the underlying infrastructure.

so , The most suitable model: **PaaS** (Platform as a Service)

justify the Answer:

This meets the team's requirements in terms of:

Focus on Coding:	PaaS provides an integrated development environment where developers can easily write, test, and deploy code.
Infrastructure Management	The service provider handles all components related to servers, storage, and the runtime environment (such as operating systems and databases). This relieves the team from worrying about managing these elements.

In comparison, IaaS (Infrastructure as a Service) requires developers to manage servers and storage themselves, adding complexity for a team that wants to focus solely on coding. On the other hand, SaaS provides ready-made solutions without the need for development, which is not suitable for developers who want to build their own applications